

The Doppler Effect for Light

Red shift and blue shift, $\Delta\lambda/\lambda = v/c$, and how the pitch of a siren tells you which way a source is moving.

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Grade 11 · Egyptian Baccalaureate

22 questions

Step-by-step solutions

$c = 3 \times 10^8 \text{ m/s}$ · $v(\text{sound}) = 340 \text{ m/s}$

PHYSICS

The toolkit for this lesson

$$\Delta\lambda / \lambda = v / c$$

the radial speed of a star or galaxy

red shift

$\lambda' > \lambda \rightarrow$ the source is moving **away**

blue shift

$\lambda' < \lambda \rightarrow$ the source is moving **towards** us

$$f' = f \cdot v / (v - v_s)$$

a sound source approaching

$$f' = f \cdot v / (v + v_s)$$

a sound source receding

c is always c

the speed of light never changes with the motion

Ask one question first: is the gap closing or opening? Closing gives a shorter wavelength — a higher pitch for sound, a blue shift for light. Opening gives a longer wavelength — a lower pitch, a red shift. Everything else in this lesson is arithmetic on top of that one idea.

Red shift and blue shift

EXPLAIN

- 1 In an experiment simulating the Doppler effect for light in the astronomy laboratory, two detectors recorded the spectral state of two stars (X) and (Y):

1. The first detector found that the spectral line of star (X) had **increased** in wavelength by an amount $\Delta\lambda$.
2. The second detector found that the spectral line of star (Y) had **decreased** in wavelength by an amount $\Delta\lambda$.

- a) Determine which of them represents a “red shift” and which represents a “blue shift”.
 b) Explain how astronomers deduce, from these spectral changes, the speed and the direction of motion of the stars in the universe.

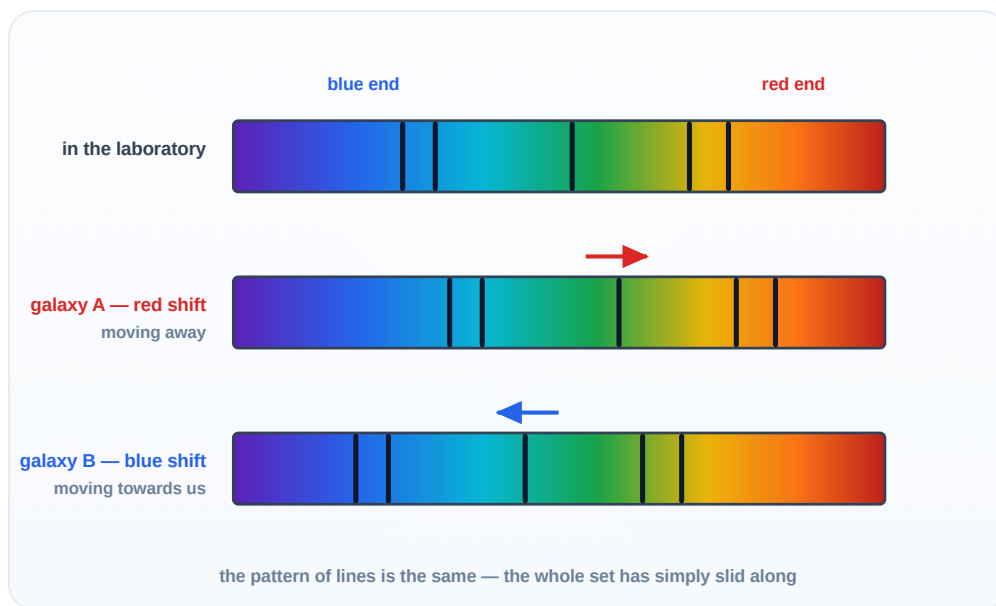


FIGURE 1

- 2 On observing light coming from a distant galaxy, a shift of its spectral lines towards the red colour is noticed. What is the physical reason for this shift, based on the Doppler effect?

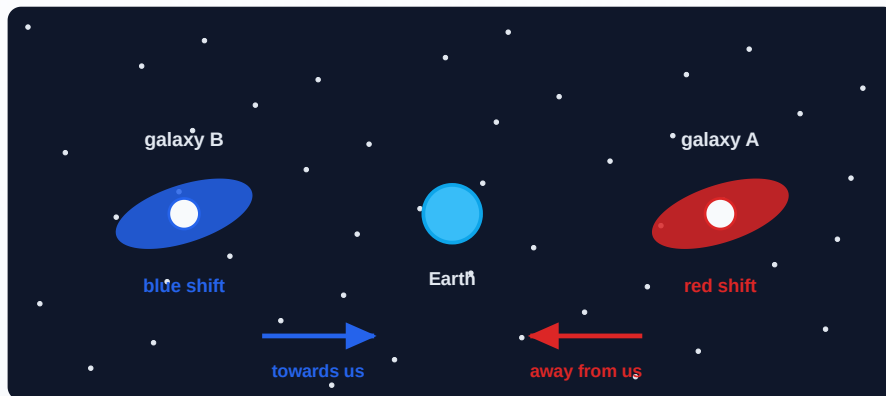
MCQ

- A the galaxy receding from the Earth leads to an increase in the wavelength
- B the galaxy approaching the Earth leads to an increase in the wavelength
- C the galaxy receding from the Earth leads to a decrease in the wavelength
- D the galaxy staying fixed in its position leads to an increase in the wavelength

Measuring the speed of a galaxy

PROBLEM

- 3 The spectrum of the element hydrogen taken in the laboratory shows a spectral line at a wavelength $\lambda = 656 \text{ nm}$. On observing the same spectral line in light coming from a moving galaxy, the magnitude of the shift in the wavelength was found to equal $\Delta\lambda = 1.312 \text{ nm}$. **Calculate the radial speed (v) of the galaxy relative to the observer**, given that the speed of light $c = 3 \times 10^8 \text{ m/s}$.



$\Delta\lambda / \lambda = v / c$ — the bigger the shift, the faster the galaxy

FIGURE 2

- 4 When a galaxy moves towards the planet Earth with a high speed, the apparent frequency of the light recorded by astronomers is:

MCQ

- A higher than the original frequency, and the lines shift towards the blue colour
- B lower than the original frequency, and the lines shift towards the red colour
- C higher than the original frequency, and the lines shift towards the red colour
- D lower than the original frequency, and the lines shift towards the blue colour

- 5 What happens to the speed of light (c) in vacuum when it is emitted from a star moving with a great speed towards the observer?

MCQ

- A it stays constant and does not change
- B it increases by an amount equal to the speed of motion of the approaching star
- C it decreases by an amount equal to the speed of motion of the approaching star
- D it changes depending on the apparent frequency of the observed light

- 6 A car moving eastwards with a speed of 15 m/s gives out a sound of frequency 600 Hz, and a motorcycle moving westwards with a speed of 20 m/s passes beside it. **Find the frequency of the sound heard by the rider of the motorcycle after they have passed each other**, taking the speed of sound in air as 340 m/s.

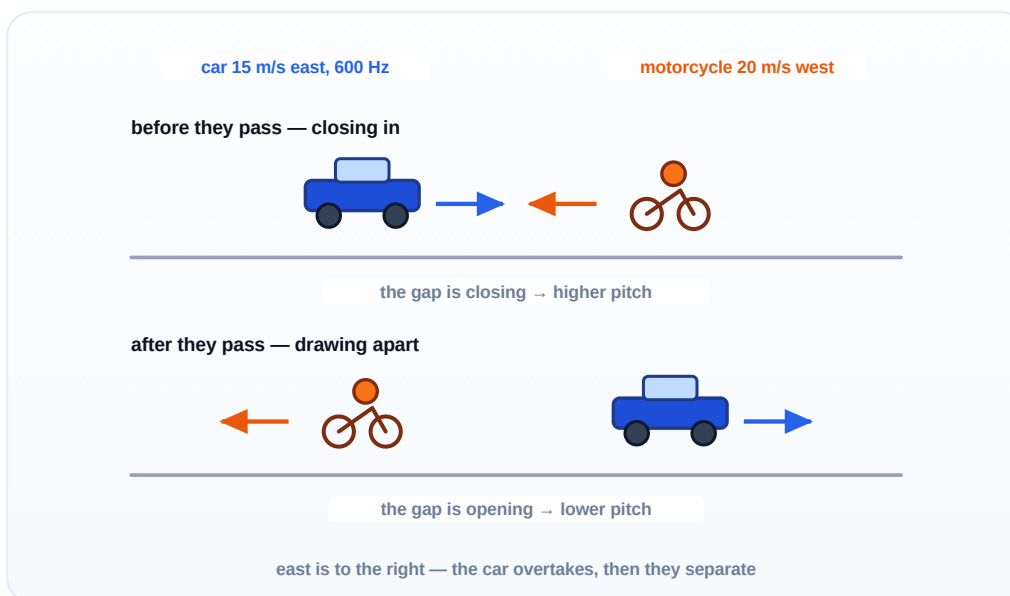


FIGURE 3

- 7 The phenomenon in which the observed frequency differs from the frequency of the source because of the relative motion between the source and the observer is called

MCQ

A interference of waves

B the Doppler effect

C diffraction of waves

D refraction of waves

- 8 An ambulance sounding its siren at a constant frequency moves with a uniform speed towards a person standing at the side of the road. What happens to the **intensity** of the sound (its loudness) and to the **pitch** of the sound (its frequency) as noticed by the stationary person while the car approaches him?

MCQ

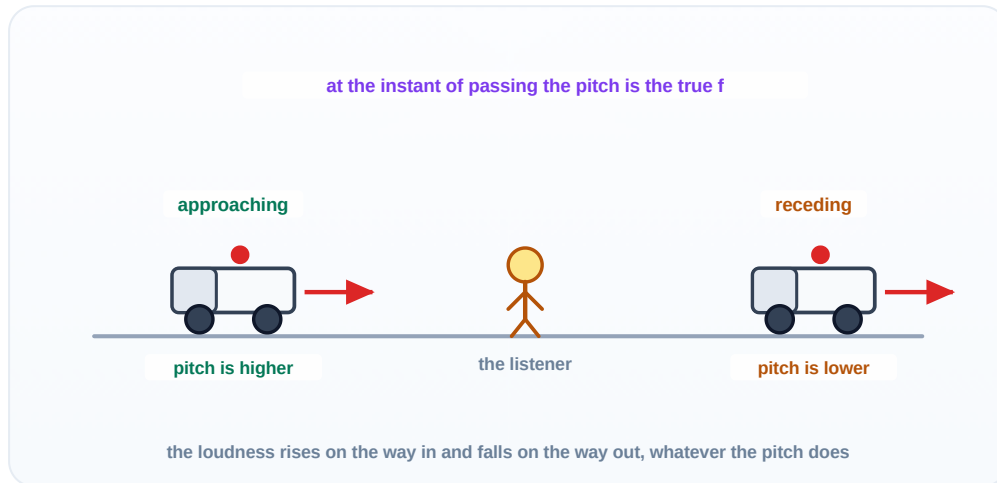


FIGURE 4

- A the intensity of the sound increases, and the apparent frequency increases
- B the intensity of the sound increases, and the apparent frequency decreases
- C the intensity of the sound decreases, and the apparent frequency increases
- D both the intensity and the frequency stay constant without change

- 9 The figure shows the observation of light coming from a galaxy; a shift of its spectral lines towards the red colour is noticed compared with the same spectrum in the laboratory. What is the physical reason for this shift, based on the Doppler effect?

MCQ

- A the galaxy receding from the Earth leads to an increase in the wavelength
- B the galaxy approaching the Earth leads to an increase in the wavelength
- C the galaxy receding from the Earth leads to a decrease in the wavelength
- D the galaxy staying fixed in its position leads to an increase in the wavelength

- 10 A sound source moving with a speed of 25 m/s approaches a stationary observer while giving out a sound of frequency 600 Hz. **Find the frequency of the sound heard by the observer**, given that the speed of sound in air is 340 m/s.

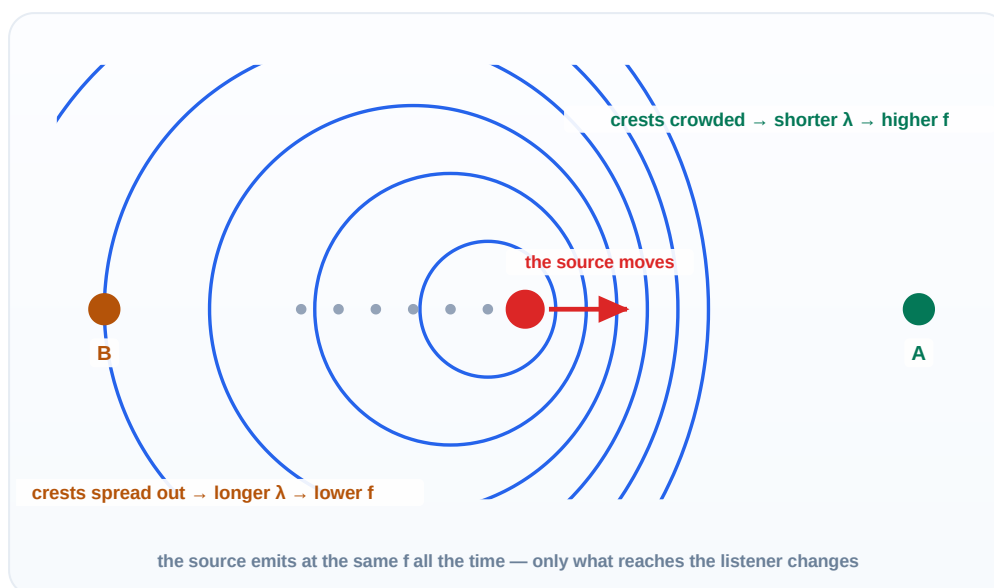


FIGURE 5

- 11 On observing the light coming from one of the distant galaxies, scientists noticed that all the colours of its spectrum were shifted slightly towards the red colour. **What do we conclude from this observation about the motion of the galaxy?** Explain the reason for this shift, based on the Doppler effect for light.

EXPLAIN

- 12 An increase is noticed in the sharpness of the whistle of a speeding train while it approaches the stopping station, while the sharpness of the sound decreases as soon as it passes the station and moves away from it. **Give the reason** for this observation, based on the concept of the Doppler effect.

EXPLAIN

WEEKLY ASSESSMENT

D Group B

- 13 When an ambulance sounding a horn of constant frequency passes a shore on which a listener is standing, and begins to move away from him with a uniform speed: how do the pitch of the sound (its apparent frequency) and the intensity of the sound measured by the listener change while the car is moving away?

MCQ

A the apparent frequency decreases, and the intensity of the sound decreases

B the apparent frequency increases, and the intensity of the sound increases

C the apparent frequency decreases, and the intensity of the sound increases

D the apparent frequency increases, and the intensity of the sound decreases

14 If the wavelength of a spectral line observed for a galaxy equals (λ') and the reference wavelength in the laboratory equals (λ), then the case that expresses the galaxy approaching with a radial speed (v) is:

MCQ

A $\lambda' < \lambda$, and the spectral lines shift towards the blue colour

B $\lambda' > \lambda$, and the spectral lines shift towards the red colour

C $\lambda' < \lambda$, and the spectral lines shift towards the red colour

D $\lambda' > \lambda$, and the spectral lines shift towards the blue colour

15 A sound source moving with a speed of 20 m/s recedes from a stationary observer while giving out a sound of frequency 600 Hz. **Find the frequency of the sound heard by the observer**, given that the speed of sound in air is 340 m/s.

PROBLEM

16 On observing the electromagnetic spectrum of light coming from a distant galaxy, one of the astronomers noticed that the absorption spectrum lines of the element sodium had shifted towards the side of the lower frequencies (the red colour). **What conclusion does the astronomer reach about the direction of motion of this galaxy relative to the Earth?** Explain your answer.

EXPLAIN

17 An ambulance approaches a person standing at the side of the road while sounding its siren, then passes him and moves away from him. **Compare** between the pitch (sharpness) of the sound that the person hears while the ambulance is approaching him, and the pitch of the sound after it has passed him and moved away.

EXPLAIN

WEEKLY ASSESSMENT

E

Group C

18 When an ambulance sounding a horn of constant frequency f approaches a person standing at the side of the road, the apparent wavelength (λ) of the sound waves that the person receives, compared with the original wavelength (λ) of the wave in air, is:

MCQ

A larger; because of the piling up of the sound waves behind the source

B smaller; because of the compression of the sound waves in front of the source

C equal; because the frequency of the sound of the source is constant and does not change

D smaller; because of the increase in the speed of propagation of sound in air

19 The spectral shift of the light of galaxies towards the longer wavelengths is known as:

MCQ

A the red shift, and it indicates that the galaxy is receding from the Earth

B the blue shift, and it indicates that the galaxy is receding from the Earth

C the red shift, and it indicates that the galaxy is approaching the Earth

D the blue shift, and it indicates that the galaxy is approaching the Earth

20 A car moving eastwards with a speed of 15 m/s gives out a sound of frequency 600 Hz, and a motorcycle moving westwards with a speed of 20 m/s passes beside it. **Find the frequency of the sound heard by the rider of the motorcycle before they have passed each other**, taking the speed of sound in air as 340 m/s.

PROBLEM

21 The siren of a fire engine is heard at a high (sharp) frequency while it approaches a person standing at the side of the road, while the same siren is heard at a low (deep) frequency after it has passed him and moved away from him. **State the name of the physical phenomenon** responsible for this observation.

EXPLAIN

22 On observing the light spectrum coming from two different galaxies (A) and (B), it was noticed that the spectral lines of galaxy (A) were shifted towards the red colour, while the spectral lines of galaxy (B) were shifted towards the blue colour. **Explain the direction of motion of each of the two galaxies** relative to the observer.

EXPLAIN



Solutions — step by step

One sentence covers the whole lesson. A **closing** gap squeezes the waves: shorter λ , higher f , blue shift. An **opening** gap stretches them: longer λ , lower f , red shift. For light, $\Delta\lambda/\lambda = v/c$ turns the size of the shift into a speed.

Part A — Classwork 2

1 the two stars X and Y

star X : λ increased $\rightarrow$ the light has been stretched $\rightarrow$ **red shift**

star Y : λ decreased $\rightarrow$ the light has been squeezed $\rightarrow$ **blue shift**

b) How astronomers use it. Every element gives out (or absorbs) light at a fixed set of wavelengths, and that pattern is measured in the laboratory. When the same pattern is found in the light of a star, the whole set of lines is seen to have slid along the spectrum without changing its **spacing** — which is what proves it is a Doppler shift rather than a different element. The **direction** of the slide gives the direction of motion: towards the red for a receding star, towards the blue for an approaching one. The **size** of the slide gives the speed, from $\Delta\lambda/\lambda = v/c$. Note that this measures only the **radial** component of the velocity — motion straight across the line of sight produces no shift at all.

	red shift	blue shift
the wavelength	$\lambda' > \lambda$ (longer)	$\lambda' < \lambda$ (shorter)
the frequency	lower than f	higher than f
the motion	the source is receding	the source is approaching
the speed from it	$v = c \Delta\lambda / \lambda$	$v = c \Delta\lambda / \lambda$

a) X shows a red shift (moving away) and Y shows a blue shift (moving towards us) b) from the size of $\Delta\lambda : v = c \Delta\lambda / \lambda$

2 the reason for a red shift

red is the **long** wavelength end of the visible spectrum, so a shift towards the red means $\lambda' > \lambda$ — the waves have been stretched out.

waves are stretched when the gap between source and observer is **opening**.

Choices (B) and (C) each get one half right and one half wrong, which is exactly what makes them tempting. Fix the pair in your memory: **red = receding = longer**.

A) the galaxy receding from the Earth leads to an increase in the wavelength

Part B — Home assignment 2

3 the radial speed of the galaxy

$$\begin{aligned}\Delta\lambda / \lambda &= v / c && \rightarrow && v = c \Delta\lambda / \lambda \\ v &= (3 \times 10^8) \times 1.312 / 656 \\ 1.312 / 656 &= 0.002 \\ v &= (3 \times 10^8) \times 0.002 \\ v &= 6 \times 10^5 \text{ m/s} = 600 \text{ km/s}\end{aligned}$$

The two wavelengths are both in nanometres, so the units cancel in the ratio and there is no need to convert either of them. Notice how small the shift is — 0.2 % — and yet it corresponds to 600 km/s, because it is being compared with the speed of light.

$$v = 6 \times 10^5 \text{ m/s} = 600 \text{ km/s}$$

4 a galaxy approaching at high speed

the gap is **closing**, so the waves are squeezed :

$$\lambda' < \lambda \quad \text{and} \quad f' = c / \lambda' > f$$

a shorter wavelength is nearer the blue end of the spectrum.

A) higher than the original frequency, and the lines shift towards the blue colour

5 the speed of light from a moving star

the speed of light in vacuum is the **same for every observer**, whatever the motion of the source or of the observer :

$$c = 3 \times 10^8 \text{ m/s} \quad \text{always}$$

the motion changes λ and f , and they change together so that $c = f \lambda$ still holds.

This is the one place where light differs completely from sound. The speed of **sound** is fixed by the air, so a wind can change it; the speed of **light** in vacuum is a constant of nature and is the foundation of the special theory of relativity.

A) it stays constant and does not change

6 the motorcycle rider after they pass

after passing, the car goes on eastwards and the motorcycle goes on westwards, so the two are **separating** : both motions lower the pitch.

$$\begin{aligned}f' &= f \times (v - v_o) / (v + v_s) \\ f' &= 600 \times (340 - 20) / (340 + 15) \\ f' &= 600 \times 320 / 355 \\ f' &= 540.8 \approx 541 \text{ Hz}\end{aligned}$$

Which sign goes where. The observer's speed is **subtracted** on the top when the observer is running away; the source's speed is **added** on the bottom when the source is running away. Both effects lower the pitch here, which is why the answer is well below 600 Hz. **Note on the printed question:** the speed of sound is not stated in this item; 340 m/s has been used, the value the book gives in the other questions of this same lesson.

$$f' \approx 541 \text{ Hz}$$

7 the name of the phenomenon

a difference between the observed frequency and the frequency of the source, caused purely by relative motion, is the **Doppler effect**.

B) the Doppler effect

Part C — Weekly assessment, Group A

8 loudness and pitch as the ambulance approaches

two separate things are happening :

the car is getting **closer** → the same energy is spread over a
smaller area → the sound is louder

the gap is **closing** → the waves are squeezed → higher f'

Keep loudness and pitch apart in your mind. Loudness depends on the **distance**, so it rises all the way in and falls all the way out. Pitch depends on whether the distance is **changing**, so it is high on the way in, true at the instant of passing, and low on the way out.

A) the intensity of the sound increases, and the apparent frequency increases

9 the red shift of a galaxy

red → longer λ → $\lambda' > \lambda$ → the gap is opening
→ the galaxy is moving away from the Earth.

A) the galaxy receding from the Earth leads to an increase in the wavelength

10 a source approaching at 25 m/s

the source is approaching, so the wavelength in front of it is shortened :

$$f' = f \times v / (v - v_s)$$

$$f' = 600 \times 340 / (340 - 25)$$

$$f' = 600 \times 340 / 315$$

$$f' = 647.6 \approx 648 \text{ Hz}$$

Check the direction before you check the arithmetic: the source is coming closer, so the answer **must** be above 600 Hz. If your calculator gives less than 600, you have put the minus sign in the wrong place.

$f' \approx 648 \text{ Hz}$

11 all the colours shifted towards the red

the whole pattern of lines has moved to longer wavelengths :

$\lambda' > \lambda$ → the gap is opening

The key point is that **all** the lines moved, and by the same **fraction** of their own wavelength. A change of the source itself would move some lines and not others; only motion slides the whole pattern in step. This is how Hubble discovered that almost every distant galaxy is receding, and that the further away a galaxy is the faster it goes — the evidence that the universe is expanding.

the galaxy is moving away from us, and the faster it recedes the bigger the shift

12 the train's whistle at the station

approaching : each crest is sent from a point nearer the station, so the crests arrive crowded → λ' is shorter → f' is higher – a sharp note

receding : each crest is sent from a point further away, so they arrive stretched out → λ' is longer → f' is lower – a deep note

The whistle itself never changes. Neither does the speed of sound in the air. The only thing that changes is the **spacing of the crests that reach the ear**.

the Doppler effect: the gap closes on the way in (higher pitch) and opens on the way out (lower pitch)

Part D — Weekly assessment, Group B

13 the ambulance moving away

the car is getting **further** → the sound is fainter

the gap is **opening** → the waves are stretched → lower f'

A) the apparent frequency decreases, and the intensity of the sound decreases

14 the case that means the galaxy is approaching

approaching → the gap is closing → the waves are squeezed :

$\lambda' < \lambda$ and a shorter λ is nearer the **blue** end

A) $\lambda' < \lambda$, and the spectral lines shift towards the blue colour

15 a source receding at 20 m/s

the source is receding, so the wavelength behind it is stretched :

$$f' = f \times v / (v + v_s)$$

$$f' = 600 \times 340 / (340 + 20)$$

$$f' = 600 \times 340 / 360$$

$$f' = 566.7 \approx 567 \text{ Hz}$$

Compare with question 10: there the source approached at 25 m/s and the pitch rose by about 48 Hz; here it recedes at 20 m/s and the pitch falls by about 33 Hz. The shift is always towards the observer's side of 600 Hz.

$f' \approx 567 \text{ Hz}$

16 the sodium lines shifted to lower frequencies

lower frequency → longer wavelength → a **red** shift

→ the gap between the galaxy and the Earth is opening.

The sodium lines are a good marker precisely because their laboratory wavelengths are known to great accuracy (the familiar orange-yellow pair at about 589 nm), so even a small displacement can be measured with confidence and converted into a speed with $\Delta\lambda/\lambda = v/c$.

the galaxy is moving away from the Earth

17 the pitch before and after the ambulance passes

approaching : the gap is closing $\rightarrow \lambda'$ shorter $\rightarrow f' > f$ (sharp)

at the instant of passing : the motion is across the line of sight $\rightarrow f' = f$

receding : the gap is opening $\rightarrow \lambda'$ longer $\rightarrow f' < f$ (deep)

The drop is heard as a sudden step rather than a slide, because the whole reversal from “closing fast” to “opening fast” happens in the fraction of a second in which the ambulance sweeps past.

the pitch is higher than the true note while approaching and lower than it after passing, and the change is sudden

Part E — Weekly assessment, Group C

18 the apparent wavelength in front of an approaching source

in the time of one cycle the source emits one crest and then moves forward before emitting the next, so the crests in front of it are closer together :

$$\lambda' = (v - v_s) / f < \lambda$$

Choice (D) contains the standard misconception. The speed of sound in air is decided by the air alone; the motion of the source changes the **spacing** of the crests, never their speed.

B) smaller; because of the compression of the sound waves in front of the source

19 the name of a shift to longer wavelengths

longer $\lambda \rightarrow$ towards the red end $\rightarrow$ a **red shift**

and stretching happens when the gap is opening $\rightarrow$ the galaxy is receding.

A) the red shift, and it indicates that the galaxy is receding from the Earth

20 the motorcycle rider before they pass

before passing, the car is still to the west of the motorcycle : the car drives east towards it and the motorcycle drives west towards the car, so the two are **closing** – both motions raise the pitch.

$$f' = f \times (v + v_o) / (v - v_s)$$

$$f' = 600 \times (340 + 20) / (340 - 15)$$

$$f' = 600 \times 360 / 325$$

$$f' = 664.6 \approx 665 \text{ Hz}$$

Put this beside question 6, which is the same pair of vehicles after they have passed: the rider hears about 665 Hz on the way in and about 541 Hz on the way out — a drop of some 124 Hz in an instant, which is why such a pass sounds so dramatic. **Note on the printed question:** the speed of sound is again not stated here, so 340 m/s has been used.

$f' \approx 665 \text{ Hz}$

21 the name of the phenomenon

the pitch changes only because the fire engine is **moving** relative to the listener – the siren itself is unaltered.

the Doppler effect

22 the two galaxies A and B

galaxy A : lines shifted towards the **red** → $\lambda' > \lambda$ → stretched
→ the gap is opening → A is moving **away** from the observer

galaxy B : lines shifted towards the **blue** → $\lambda' < \lambda$ → squeezed
→ the gap is closing → B is moving **towards** the observer

Almost every galaxy we can see shows a red shift, so a blue-shifted galaxy such as B is unusual and means it is close enough for its own motion within the local group to beat the general expansion. The Andromeda galaxy is the famous example — it is blue-shifted, and is approaching us at about 110 km/s.

A is receding from the observer and B is approaching the observer